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Does Transparency in Indian IPOs Help Retail Investors?

Abstract

Indian regulations for institutional investors' participation in the IPO book building process are unique when compared to the developed nations. Institutional investors can invest in two stages in the Indian book building process. As anchor investors before the issue opens to the public and as qualified institutional investors (QIBs) during the book building phase. Investment banker has the discretionary power to allocate shares to the anchor investors like developed countries book building process, but the details like price, identity of anchor investors and quantity allocated is revealed before the issue opens to public. On the other hand, during the book built period, the subscription pattern of all the investors is made live to public on real time basis. The study tries to investigate if the transparent way in which the bidding information of institutional players are made public has any influence on retail investors subscription and short term IPO performance. The study also tries to find out the anchor specific characteristics which influence the subscription of investors.

The results show that anchor investors play a certification role for institutional investors only. Both institutional and retail investors observe and follow each other's bids during the book building period. High institutional and retail subscriptions have positive effect on initial returns. In anchor backed IPOs large number of business conglomerate participation and anchor oversubscription has high positive impact on both subscription and initial returns.

Key Terms: IPO, Transparency, Book Building Process, Anchor Investors, Institutional Investors, Retail Investors, Oversubscription, Underpricing

1. Introduction

Out of the different parties involved in the IPO process, few are more informed than the rest. This gives rise to information asymmetry, which sometimes is so high that even the issuer and underwriter are not confident of the company's valuation and rely on the institutional investors to guide them about the right pricing of the issue (Rock, 1986). The institutional investors which are more informed than the rest (Chakravarty, 2001; Field, 2009 and Titman, 2006**)** generally use value based approach in stock selection and invest in undervalued stocks (Nofsinger and Sias, 1999; Chemmanur, He, and Hu, 2009). The underwriter tries to extract information regarding the true value of the issue from the institutional investors and in a quid pro quo arrangement underprices the issue (Benveniste and Spindt, 1989; Hanley and Wilhelm, 1995). Allocation is also at the discretion of the underwriters and they tend to favor their buy side clients and the ones whose bids are more informative than the rest (Cornelli and Goldreich, 2001; Loughran and Ritter, 2002; Pulliam and Smith, 2000, 2001). On the other hand, due to limited access of relevant information and lack of expertise to select good IPO companies the uninformed investor bids haphazardly (Derrien 2005; Ljungqvist et al., 2006) and tend to subscribe the overpriced issues. (Amihud et al., 2003) analyzed Israeli IPOs and found that uninformed investors earn negative allocation weighted initial returns. Sometimes, uninformed investors also rely on certain certifications like presence of venture capitalists (Megginson and Weiss, 1991) underwriter's reputation (Carter et al., 1998) and in Indian context IPO grading (Deb and Marisetty, 2010) to select good IPOs. But most of these certifications have proved to be of limited use (Chemmanur and Krishnan, 2012) and do not help in the alleviation of information asymmetry. The documented studies by (Lee and Wahal, 2004; Lougran and Ritter, 2002 and Khanna et al., 2008, Liu and Ritter, 2010) highlight that issuing firms gives several incentives to the certifying agencies and raised doubts about the quality of the certification they provide. If the uninformed investors can get to follow the investment pattern of the informed counterparts then the problem of asymmetry can be possibly alleviated. (Amihud et al., 2003) also suggested that in order to reduce their losses uninformed investors should follow publicly available information. In developed countries the book building exercise is very secretive and the allocation data is not revealed by the underwriter and investors at best can access only analysts, brokerage houses or investment bankers' reports.

On the other hand, Indian IPO process is unique and transparent. In India institutional investors bid at two stages. The first stage is a day before the IPO open for public bidding and the second stage is during the book built process. In the first stage the bids are collected from the institutional investors, also known as anchor investors, and all the relevant information like identity of anchor investors, allocation price and quantity is made public, before the commencement of the second stage. During the second stage, all the bidding information is made live. The transparency in Indian IPO process enables all the investor categories to follow each others' bidding details. Using anchor and daily book built data; the present study tries to find out the effect of early bids from informed investors (both anchor investors and non anchor qualified institutional investors, also known as QIBs) on their uninformed counterparts i.e. retail investors and underpricing.

In the IPO literature there are a number of studies in connection with certification effects of investment bankers' reputation and venture capital investment but the studies on cornerstone investors (anchor investors in Indian context) are very few. This is mainly because cornerstone or anchor investors are generally not common in the investor community of the developed nations. This study also tries to add to the IPO literature the certification effect of anchor investors and their specific characteristics which can have high implication on subscription and underpricing. Anchor identity, affiliation with investment banker, price of investment, oversubscription, foreign allocation etc are few distinctive anchor features considered for the same.

The results of the study show that anchor investors backed IPOs are more oversubscribed and anchor investors influence the institutional investors' subscription as compare to retail investors. Unlike (Khurshed et al., 2014) the results of the study show that both institutional and retail investors influence each other's subscription and the information flow is not unidirectional from institutional to retail investors. However it support (Khurshed et al., 2014) finding that retail investors submit their bids late in the book built period. Analysis of the anchor data reveals high anchor oversubscription and large number of business group participation leads to high institutional subscription in book building phase. High oversubscription in all the categories i.e. anchor, institutional and retail investors has a strong positive effect on short term IPO performance.

The paper is organized in different sections. Section 2, discusses transparent IPO book building process in India and anchor investors in book building. Section 3 defines the testable hypotheses. Section 4 describes the data and methodology. Section 5 presents the results and Section 6 provides the conclusion.

2.1 Transparent book building Process

For efficient price discovery of IPOs, book building process is used worldwide. India has a unique transparent book building process similar to open auction, where the demand can be observed on a real time basis during the time period the book is built.
The issuer planning to raise money from public using book building appoints the lead merchant bankers as 'book runners'. In consultation with the lead manager the number of securities to be issued and price band is finalized. After that syndicate members with whom the bids are placed by the investors are appointed. The bids are entered by book-runners and the syndicate members online on real-time basis for the entire bidding period into a software called 'electronic book'. Both National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) provide the facility of maintaining electronic book by the book runners and syndicate members. During the book built period, the bids are collected from investors (QIBs, non institutional investors and retail investors which bid for their respective reserved quota) at different prices, within the price band specified by the issuer. The details of the bids are made public on a real time basis. Bids can be revised by the bidders before the book closes. The book normally remains open for a period of 5 days. After the book building period is over, the book runners assess the demand of the bids at different price levels. The book runners and the issuer decide the final offer price. Generally, the number of shares to be issued is fixed and the issue size is determined depending on the final offer price per share. Allocation of shares is done to the successful bidders and to the rest refund is given.

2.2 Anchor Investors in Book Building Process

In order to help market participants market the issues in an uncertain market, Securities and Exchange Board of India (SEBI) in July 2009, introduced a new category of investors, anchor investors. Anchor investors are institutional investors like mutual funds, insurance companies, pension funds, foreign funds that invest in shares before the IPO opens for public subscription.

In India, book built IPOs have a 50 per cent reservation for qualified institutional buyers (QIBs). Out of this 50% quota reserved for QIBs, an issuer is allowed to allot up to 60%[1] of the portion to anchor investors i.e. 30 per cent of the total issue size. One-third of the anchor investors' quota is reserved for domestic mutual funds. Institutional investors applying in anchor category get a firm allotment while in the public offer the allotment to institutional investors (QIBs) depend on the number of times the issue gets subscribe. Anchor investors are required to pay a margin of at least 25% on application, and the balance is paid within two days of the date of closure of the issue. On the other hand QIBs are required to pay a minimum margin of 10%. There is a lock in period of 30 days for anchor investors from date of allotment while QIBs can sell on listing day.

Anchor participation in public issues is optional and the minimum bid for them is Rs 10 crores in an IPO. In case the issue size is up to Rs 10 crores, maximum of two anchor investors can get allotment. For issues above 10 crore and up to Rs. 250 crore, minimum of two and maximum of fifteen anchor investors can get allotment, subject to minimum allotment of Rs. 5 crore per anchor investor, for issues above 250 crores[2] there can be 10 additional investors for every additional allocation of Rs.250 crore, with minimum allotment of Rs.5 crore per anchor investor.

Anchor investors bid one day before the issue opens for public and the allotment process is completed on the same day. Generally the negotiations with prospective investors take place before the bidding. Allocation details like quantity and price is also made public on the same day i.e. a day before the issue opens for public. Since the final issue price is determined after the culmination of book building process and the anchor investors are allotted shares before that, there can be a price differential between anchor investors price at which the shares are allotted to them and the price at which the other investors like QIBs, retail investors and non institutional investors are allotted shares. If the price determined through book building is lower than the price at which anchor investors are allotted shares, then the surplus amount is not refunded to anchor investors. However, if it is higher, then the anchor investors are required to pay the difference.

Allocation of shares to anchor investors is done on a discretionary basis by the investment banker like in case of developed countries book building process. However, the selection criteria of anchor investor has to be well defined by the investment banker and made available for SEBI's inspection.

1. Prior to June 2014, up to 30% of the QIB portion in an issue was reserved for anchor investors. On June 19, 2014 SEBI increased the reserved quota from 30% to 60%.
2. In September 2015, SEBI removed the restriction on the maximum number of anchor investors (earlier it was 25) for anchor allocation of public issue worth over Rs.250 crore.4

3. Hypotheses

The present study evaluates the significance of globally documented certification mechanisms like investment bankers' reputation, venture capital backing on the subscription level and performance of book built IPOs. It also checks the effect of IPO grading which is unique to Indian IPOs. The study tests if the timely information of bidding pattern of the institutional investors has any significant impact on retail investors' and underpricing. The study also evaluates anchor investors participation and their effect on subscription and IPO performance.

Beatty and Ritter, 1986; Carter and Manaster , 1990; Maksimovic and Unal , 1993; Cater, Dark and Singh, 1998 and recently in India, Sahoo and Rajib, 2010 found that IPOs associated with reputed investment bankers had low underpricing. The study hypothesizes that the presence of reputed investment bankers should increase subscription and decrease underpricing.

Venture capitalists believed to invest in companies with strong management team and high growth potential. Therefore the companies backed by venture capital funds are expected to have strong fundamentals and corporate governance. Barry, Muscarella and Vetsuypens, 1990 and Megginson and Weiss, 1991 found lower underpricing in IPOs with venture capital funding in pre IPO period. But Francis and Hasan, 2001; Franzke, 2001 and Lee and Wahal, 2004 found higher underpricing in venture capitalist backed IPOs. The present study remains ambiguous on the effect of venture capitalist certification role.

IPO grading was made mandatory by SEBI in May 2007 to protect retail investors' interest but the same was made voluntary in Dec 2013 to reduce the reliance on Credit Rating Agencies [3]. Deb and Marisetty, 2010 found that firms with higher grades had lower underpricing, Khurshed et al., 2014 documented that grading does not affect underpricing, while Jacob and Agarwalla, 2012 showed high grade firms were associated with higher underpricing. Considering the past documentation, the present study is ambiguous on the impact of IPO grading on subscription and underpricing.

As discussed earlier, In India, institutional investors participate as anchor investors and qualified institutional investors (QIBs) in Indian IPOs. Anchor investors bidding information is made available to retail investors, before they start bidding for their quota. And bidding details of QIBs are available on a real time basis to retail investors during the book built period. The present study aims to see the effect of the early or live institutional bidding on retail investors' subscription, as before making their investment decision retail investors know where their informed counterparts are investing and get enough opportunity to follow them. Neupane and Poshakwale, 2012 and Khurshed et. al, 2014 also studied Indian transparent book building process and showed that the participation of retail investors is significantly influenced by the participation of institutional investors. Bubna and Prabhala, 2013 found anchor investors participation influence only the institutional investors bidding. The study hypothesizes a high positive effect of early institutional bids on retail subscription.

Bubna and Prabhala, 2013 found that there is substantial heterogeneity in the implementation of anchor backed IPOs. Underpricing in anchor-backed IPOs decreases with fewer unique investor families and when anchor and underwriters are connected. The present study assumes that large numbers of business group participation, high allocation of shares to foreign investors, high anchor subscription, bidding at the upper end of the price band, and full allotment of the reserved shares, all indicate towards high value of an IPO. The study hypothesize that all these variables should have a positive effect on the overall subscription.

4. Data

The sample includes all the 162 bookbuilt IPOs on BSE and NSE from July 2009 to March 2016. Out of these 162 IPOs, 78 IPOs are anchor investor backed [4].

The data for the study is collected from various sources. The key data on firm and IPO characteristics like on age of the firm, issue size, IPO grading, venture funding in pre IPO period, names of investment bankers are all collected from the IPO prospectus which are available on the market regulator SEBI's website. Market data is collected from the BSE/NSE website. BSE Sensex index is used as the market index to calculate the market adjusted initial returns. Data on IPO subscription is obtained from the BSE and NSE websites and from a primary IPO investment portal Chittorgarh[5]. Data on anchor investors is collected from BSE/NSE website, PRIME database, few finance portals like Money Control[6], and business news portals like Economic Times[7], Hindu Business Line[8], Business Standard[9], etc.

4.1 Variables Used for Certification Effect

Following Equations 1 and 2 check the effects of different certification mechanisms on the subscription patterns of institutional and retail investors.

$$\text{Total Subscription QIB} = \alpha + \beta 1 \text{ Anchor Investor} + \beta 2 \text{ IB Reputation} + \beta 3 \text{ VC backing} + \beta 4 \text{ IPO Grade} + \beta 5 \text{ Penultimate Subscription RI} + \beta 6 \text{ Age} + \beta 7 \text{ Issue Size} + \beta 8 \text{ Market Condition} + \beta 9 \text{ IPO Year} + \varepsilon. \quad (1)$$

$$\text{Total Subscription Retail} = \alpha + \beta 1 \text{ Anchor Investor} + \beta 2 \text{ IB Reputation} + \beta 3 \text{ VC backing} + \beta 4 \text{ IPO Grade} + \beta 5 \text{ Penultimate Subscription QIB} + \beta 6 \text{ Age} + \beta 7 \text{ Issue Size} + \beta 8 \text{ Market Condition} + \beta 9 \text{ IPO Year} + \varepsilon. \quad (2)$$

In Eq1, Total Subscription QIB is measured as the ratio of the total shares subscribed by the institutional investors to the total number of shares available to QIBs for allotment. Anchor investor is a dummy variable which takes a value of 1 if in an IPO anchor investors have participated and 0 otherwise. IB Reputation is a measure of the reputation of the investment banker and takes a value of 1 if there is a minimum of one investment banker in the top five rank by market share of total IPO mobilization in the IPO year, and zero otherwise. VC backing is a dummy variable that takes a value of 1 if an IPO is Venture Capitalists and Private equity investors backed and 0 otherwise. IPO Grade is the grade awarded to the IPO firm by a grading agency or 0 for non-graded IPOs.

3. "SEBI makes IPO grading mechanism voluntary", The Hindu (Dec 24,2013). Available athttp://www.thehindu.com/business/Industry/sebi-makes-ipo-grading-mechanism-voluntary/article5498323.ece

4. The study extends the empirical evidence on anchor backed IPOs, previously studied by Bubna and Prabhala, 2013 on a sample of 49 IPOs on the BSE and NSE between 2009 to 2012.

5. www.chittorgarh.com

6. www.moneycontrol.com/

7. economictimes.indiatimes.com/

8. www.thehindubusinessline.com

9. www.business-standard.com

Penultimate subscription RI is the ratio of the total shares subscribed at the end of the penultimate day of book building to the total number of shares available to retail investors. The present study takes the index return of thirty days before the IPO opening as a proxy for market condition. There are also some control variables like age of the IPO firm, issue size and year dummies.

In Eq2, Total Subscription Retail is measured as the ratio of the total shares subscribed by the retail investors to the total number of shares available to them. The explanatory variables are the same as the first equation except for Penultimate Subscription QIB which is the ratio of the subscriptions by institutional investors at the end of the penultimate day of the book building process to the total number of shares available to them.

In the equation 2, if the coefficient of anchor investor and penultimate QIB variables are positive and significant, then it will show that retail investors gets influenced by the institutional bidding.

The same certification variables are used to check the effect of certification on underpricing :

Underpricing = α + β1 Anchor Investor+ β2 IB Reputation+ β3 VC backing+ β4 IPO Grade + β5 Subscription + β6 Age+ β7 Issue Size + β8 Market Condition + β9 IPO Year + ε. (3)

In Eq3, Subscription is the ratio of the total number of shares subscribed by each category (QIB, non institutional investors and retail investors or Total) of investors to the total number of shares available to them. Underpricing is the market adjusted difference between the offer price and the first-day closing price of an IPO.

4.2 Anchor Specific Variables

Following Equations 4 and 5 checks the effect of anchor specific features on the subscription patterns of institutional and retail investors in anchor backed IPOs[10].

Subscription QIB = α + β1 Total No. of Business Group+ β2 Foreign Allocation+ β3 Investor Banker Affiliation+ β4 Subscription at Upper Limit of Band + β5 Subscription to maximum limit + β6 Oversubscription_Anchor+ β7 Age +β8 Issue Size + β9 Market Condition + β10 IPO Year + ε. (4)

Subscription RI = α + β1 Total No. of Business Group+ β2 Foreign Allocation+ β3 Investor Banker Affiliation+ β4 Subscription at Upper Limit of Band + β5 Subscription to maximum limit + β6 Oversubscription_Anchor+ β7 Age +β8 Issue Size + β9 Market Condition + β10 IPO Year + ε. (5)

In Eq4, Subscription QIB is the ratio of total shares subscribed by the QIBs to the total number of shares available to them in the anchor backed IPOs and in Eq5, Subscription RI is the ratio of total shares subscribed by the retail investors to the total number of shares available to them in the anchor backed IPOs

Anchor investors can be classified on the basis of their origin, business group, affiliation with investment banker and the price at which they make their investments. Total No. of Business Group is the number of unique business groups participating as anchor investors. Anchor investors can be classified according to the business group they belong to i.e. if anchor investors belong to same business group apply through different applications in an IPO then they are classified as one group. E.g., if Birla Sun Life Mutual Fund and Birla Sun Life Insurance (which are part of Birla group) apply through different applications then their application is considered as one. A large number of unique groups participating in anchor round signal a high demand by varied institutions. The present study label each anchor as foreign or domestic. Foreign Allocations are the percentage of total equity shares allocated to the foreign anchor investors.

If anchor investors are from the same business group as investment banker then there is chances that investment banker may use his discretionary power in share allocation to them or benefit the group company by higher underpricing. On the contrary use the information generated at the group level to set the offer price closer to its true value (Bubna, Prahala, 2013). Investor Banker Affiliation is a dummy variable that takes a value of 1 if anchor investor(s) is affiliated to the investment banker(s) of the issue and 0 otherwise.

Anchor investors are required to pay higher of the price determined during the two stages (anchor stage and book building stage) within the price band. If anchor investors subscribe at the upper band in the first stage itself then it shows their confidence in the quality of the IPO. Subscription at Upper Limit of Band is a dummy variable which takes a value of 1 if the subscription of shares by anchor investors is made at the upper end of the price band and 0 otherwise.

If anchor investors fully subscribe their reserved quota and have high subscription levels then it indicates their high confidence in the IPO. Subscription to maximum limit is a dummy variable which takes a value of 1 if the anchor investors are allotted their entire reserved quota of shares and 0 otherwise. Anchor_Oversubscription is measured as the ratio of the total shares subscribed by the anchor investors to the total number of shares available to them.

In the above two equations if a coefficient of a variable is positive and significant, then the specific variable will influence institutional (Eq4) or retail (Eq5) subscription.

The same above variables (in Eq4 or Eq5) are used to check the effect of anchor specific characteristics on the underpricing

Underpricing = α + β1 Total No. of Business Group+ β2 Foreign Allocation+ β3 Investor Banker Affiliation+ β4 Subscription at Upper Limit of Band + β5 Subscription to maximum limit + β6 Oversubscription_Anchor+ β7 Age +β8 Issue Size + β9 Market Condition + β10 IPO Year + ε. (6)

Here Underpricing is the market adjusted difference between the offer price and the first-day closing price of an anchor backed IPO. If a coefficient of a variable is negative and significant, the specific variable will decrease underpricing.

10 IPOs in which anchor investors had participated

5. Results

5.1 Descriptive Analysis

Table 1 shows the IPO activity in the sample period July 2009- March 2016. A total of 162 firms went public during the time period and anchor investors invested in 78 firms. The table also highlights the decrease in number of graded IPOs after it was made voluntary in Dec 2013.

Table 2 presents descriptive statistics of the key characteristics of issuing firms. The mean (median) IPO issue size is Rs 518.31(186.65) crs. Consistent with Bubna and Prabhala, 2013 issue size of anchor backed IPOs are significantly greater than the issue size of non anchor backed IPOs and highlights requirement of more categories of investors in bigger issues. Mean (median) underpricing during the study of the period is 8.95 %(3.2%) and the difference between Anchor backed and non anchor backed firms underpricing is not significant. Hence the preliminary analysis shows that presence of anchor investors has no significant impact on reducing underpricing.

Anchor-backed IPOs are associated with significantly higher number of reputed investment bankers and percentage of VC backing as compared to non-anchor IPOs. The mean (median) age of anchor firms is 17.447(13.003) years in comparison to 16 (12.36) years in case of non anchor IPO firms. The difference in age and market condition is not significant.

The median (mean) anchor-backed IPOs total subscription is 16.43 (6.31) compared to 8.885 (2.21) in non anchor backed IPOs, showing anchor backed IPOs are significantly more oversubscribed and anchor investors do play a certification role in attracting investors. Again consistent with Bubna and Prabhala, 2013, higher subscription in anchor-backed IPOs is primarily driven by institutional investors. The institutional category is over subscribed by mean (median) 18.036(7.446) times compared to 8.342(.86) for non anchor IPOs. However, the retail subscription is significantly higher in case of non-anchor IPOs. This is same in all the other days of the book building period. Thus, presence of anchor investors has a certification effect only on institutional investors and not on the retail investors. Table 2 also highlights that during the entire duration of book building process i.e. on first day, first day to penultimate day, penultimate day and last day institutional subscriptions are higher than retail subscription.

Table 3 shows the descriptive statistics of key IPO backed characteristics. Foreign anchor investors actively participate in Indian IPOs and on average, accounts for 53% of the total equity shares allocated to anchor investors. This shows they want to get assured subscription before the issue is made public. The results support that a large number of unique business group participates as anchor investors. On average, the allotment of the shares is done to eight different groups in an IPO. Generally companies of the same family as underwriter refrain to invest. Only 17% of anchor investors belong to the same business group as investment banker. Mostly anchor investors invest in those companies about which they confident as a majority (72%) of the IPOs are subscribed at the upper end of the limit. In 85% IPOs anchor investors fully subscribe their percentage of reserved quota from QIB subscription. And on average, anchor tranche is oversubscribed by 1.478 times.

5.2 Multivariate Analysis

Table 4, considers all the book built IPOs between July 2009 and March 2016. Model 1 and 2 tests total subscriptions of institutional and retail investors as dependent variable. Model 1, shows that presence of anchor investors and IPO grading both have high certification value for institutional investors. As documented by Bubna and Prahala, 2013, IPOs backed by anchor investors have high institutional subscription. In case of Retail investors, the presence of anchor investors is associated with lower significant total subscription. IPO grading has a positive impact on retail investors (Model 2). Apart from the different certification mechanisms, the effect of transparency during book building phase is also checked. For this the penultimate demand of retail and institutional investors are included as explanatory variables. The results show that the transparency in the book building process influences the decisions of both the investors. Not only the retail investors observe and follow the early bids of institutional investors but institutional investors also follow the bids of the retail investors. This is in contrary to Khurshed et. al, 2014 study, according to which early bids from retail investors do not affect institutional bids, whereas early institutional bids have a strong positive impact on retail bids. This can be due to the increase in investments by the institutional investors in those issues in which they see high retail interest, so that they can hype the issue and can make money by flipping the shares post listing Aggarwal, 2003; Chemmanur, He, and Hu, 2009. Thus the early bids from one category of investor have a strong positive impact on the total subscription of the other category.

Table 4 also evaluates the influence of the certifications at different points in time during the book building process. For this the daily demand buildup of both the category of investors is considered. None of the certification mechanisms plays any significant role in influencing the demand of either of the investors on the first day (Model 3 and 4). Model 5 and 6 tests demand build up from first to penultimate for institutional and retail investors. Only IPO grading has the positive effect during this period. Last day of the book building process which sees the highest subscriptions in both the categories (refer, Table 2) anchor backed IPOs sees more institutional subscriptions (Model 7, Table 4). Penultimate demand of the retail investors is the other significant influencing variable. On the other hand, IPO grading and high penultimate day QIB subscription leads the retail demand buildup from penultimate day to the last day of the book (Model 8). This again is contrary to Khurshed et. al, 2014 findings, according to which grading does not provide any additional information to retail investors other than what is already provided by the transparency of the book building process.

Table 5 exhibits the impact of certification mechanism and subscription on underpricing. Model 1 and 2, tests the effect of certification mechanism on underpricing. Models 3-5, considers all the certification mechanisms along with subscription level of institutional and retail investors. The coefficient of anchor investors is insignificant in all the models and it shows that anchor backing does not help in reducing information asymmetry. VC backing also does not have a significant effect on underpricing. Investment banker's reputation seems to have a certification role in reducing information asymmetry, although in model 3, the coefficient is significant only at the 10% level. Institutional, retail and total subscription all have a high positive impact on underpricing, in accordance with Cornelli and Goldreich, 2003, Bubna and Prabhala, 2011 and Khurshed et al., 2014.

Anchor investors can be classified on the basis of their origin, business group, affiliation with investment banker and the price at which they make their investments. The study test the effect of these characteristics on subscription (Table 6) and underpricing (Table7).

Model 1 and Model 4 in Table 6 takes into account the effect of anchor identity on institutional and retail investors' subscription. Model 2 and 5 takes into consideration the investment decisions of anchor investors in terms of price and oversubscription on institutional and retail investors. Model 3 and 6 includes all the variables and tests the effect on institutional and retail investors' subscription. Higher number of unique business groups participating in the anchor round has a high positive significance on institutional subscription. The variable shows positive effect on retail subscription in Model 4 but is insignificant in full model (Model 6). Higher allocation of shares to foreign investors should increase the demand of the investors but institutional investors view it with caution and their subscriptions decreases in the issues which have higher percentage of shares allocated to foreign investors. The effect is same in case of retail investors, although the coefficient is significant only at 10% in full model. The relationship between the investment banker and the anchor investors participating in the first stage is insignificant for institutional investors but is highly significant to retail investors and affect their subscription decision. Anchor investors bidding at the upper limit of the band communicate their confidence in the IPO and hence positively affect institutional bidding but are insignificant in case of retail bidding. On the contrary anchor investors full subscription of their reserved quota has a higher significant impact on retail bidding as compare to institutional bidding. The number of times anchor tranche is oversubscribed has a significantly high positive effect on both institutional and retail subscription.

Table 7 shows the implication of anchor characteristics on underpricing in anchor backed IPOs. Model 1 test the effect of anchor identity and Model 2 accounts for effect of anchor investors pricing and oversubscription on underpricing. Model 3 include all the anchor characteristics and check their effect on underpricing. Table 7 shows that a large number of business group participation and anchor oversubscription both have a high significant positive effect on underpricing. Anchor bidding at upper band is significant in Model 2 but is insignificant in full model. Other anchor characteristics do not have any impact on underpricing.

6. Conclusion

India has a unique two tier transparent book building process. In the first stage, only the more informed investors participate for their reserved quota. At this stage the allocation power is with the investment banker but the bidding details are made public before the second stage begins. And in the second stage investors can monitor each others' bid on a real time basis and can also revise their bids before the book closes. The study checks the relevance of this two level transparency in the book building process on the subscription pattern of the retail investors and on short term market performance. The results of the study show that anchor investors who participate in the first stage have a high positive significant influence only on the institutional investors. Anchor backed issues see low retail demand. But overall anchor backed IPOs are more oversubscribed than non anchor backed ones. Anchor issues which have high anchor over subscription also see high underpricing.

During bookbuilding, institutional investors make early bids than retail investors but the flow of information as earlier documented (Khurshed et. al, 2014) is not unidirectional. Both institutional and retail investors significantly influence each other's demand. IPOs which have high oversubscription also see high underpricing. The findings suggest that institutional investor's using their own informational advantage and by following anchor subscription (in case the issue is anchor backed), bid early in the book building period. This has a signaling effect on retail investors and they follow their more informed counterparts' investment. Meanwhile, institutional investors also keep a track of retail bids and increase subscriptions in the issues which gather high retail interest. Thus the issue gets oversubscribed and hyped. This gives an opportunity to institutional investors to easily flip the shares of the hyped issue in the initial trading days and make profit.

The study also checks the effect of certification mechanism on the subscription and underpricng. The results show that internationally documented certifications like investment banker's reputation and VC presence in the pre issue funding period has no influence on the subscription of either category of the investors. IPO grading has a strong positive impact on subscription especially in case of retail investors. This further suggest the fact that institutional investors follow anchor investors and retail investors in turn follow QIBs, so in future with voluntary IPO grading (which had led to decrease in number of companies getting IPO grading, Table 1) and increase in number of anchor backed issues (since April 2015, Table 1) anchor backing directly or indirectly can be one of the most influential certification mechanism for the investors.

The present study, which also tries to contribute to the IPO literature in terms of certification effect of anchor investors, finds that large numbers of unique business group participation and anchor oversubscription, both signaling high anchor demand and their high valuation of the IPO, leads to increase in institutional investors demand and IPO underpricing.

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Table 1: Number of IPOs by year. The table presents the count of IPOs in India between July 2009 and March 2016.

IPO Year	Total Book Built IPOs	Non Anchor IPOs	Anchor IPOs	Graded IPOs	Non Graded IPOs
July 2009- March 2010	38	22	16	38	0
April 2010- March 2011	50	31	19	50	0
April 2011- March 2012	32	26	6	32	0
April 2012- March 2013	9	1	8	8	1
April 2013- March 2014	1	0	1	1	0
April 2014- March 2015	8	0	8	2	6
April 2015- March 2016	24	4	20	1	23
Total	162	84	78	132	30

Table2: Descriptive statistics of anchor-backed, non-anchor backed and all book built IPOs between July 2009- March 2016. Issue size is the issue proceeds (in Rs. Crs.). Underpricing is the market adjusted difference between the offer price and the first-day closing price of an IPO. Total Subscription is the number of times issue is subscribed. It is listed separately for different investor categories (QIB and Retail) and different points in time during the book building process (first day, demand build up from first day to penultimate day, penultimate day and last day). IB Reputation in an IPO takes value 1 if there is at least 1 lead manager in the top 5 rank by share of total IPO proceeds in the IPO year, and zero otherwise. VC backing is a dummy variable that takes a value of 1 if an IPO is VC backed and 0 otherwise. Age is the difference between the firm's IPO year and its incorporation year. Market condition is the index return of thirty days before the IPO opening. The last column reports the significance of the difference in variables means for anchor-backed and non-anchor backed IPOs.∗∗∗, ∗∗ and ∗ denote significance at the 1%, 5% and 10% levels respectively.

	Obs	Mean	Median	Min	Max	Obs	Mean	Median	Min	Max	Obs	Mean	Median	Min	Max	Diff
	Anchor					Non Anchor					All IPOs					
Issue Size(Crs.)	78	592.2	400	35	4155.8	84	448.02	77.44	1.5	15199.44	162	518.31	186.65	1.5	15199.44	***
Underpricing (%)	78	10.04%	4.50%	-38.59%	96.49%	84	7.91%	1.19%	-75.09%	144.52%	162	8.95%	3.20%	-75.09%	144.52%	
IPO Grading	78	3.519	4	2	5	84	2.375	2	1	5	162	2.317	2	0	5	
Total Subs QIB (times)	78	18.036	7.449	0.68	93.86	84	8.342	0.86	0	144.428	162	13.0695	2.28	0	144.428	***
Total Subs Retail (times)	78	4.55	1.979	0.08	41.26	84	6.1431	3.4	0.1604	44.45	162	5.366	2.853	0.08	44.45	**
Total Subs (times)	78	16.43	6.31	0.7575	86.33	84	8.885	2.21	0.77	93.6	162	12.5631	2.9612	0.7575	93.6	**
First Day subs QIB (times)	78	0.6402	0.19	0	10.864	84	0.4601	0.18	0	6.0057	162	0.548	0.18	0	10.8642	
First Day Subs Retail (times)	78	0.21859	0.1	0	2.7	84	0.39672	0.1845	0	2.43	162	0.3099	0.1204	0	2.7	**
Penultimate Subs QIB (times)	78	4.383	0.95	0	47.45	84	2.552	0.47	0	49.8	162	3.445	0.6276	0	49.8	**
Penultimate Subs Retail (times)	78	0.9432	0.4	0.01	12.94	84	1.1229	0.68	0.0488	10.86	162	1.0353	0.5542	0.01	12.94	*
Last Day subs QIB (times)	78	13.65	4.22	0	87.39	84	5.79	0.16	-0.2886	134.7	162	9.625	0.7872	-0.2886	134.7	***
Last Day Subs Retail (times)	78	3.607	1.44	0.0469	28.32	84	5.02	2.66	0.09	36.07	162	4.331	2.135	0.0469	36.07	*
First Day to Penultimate Day QIB(times)	78	3.743	0.64	0	47.45	84	2.092	0.1831	-0.05	48.13	162	2.897	0.29	-0.05	48.13	**
First Day to Penultimate Day Retail (times)	78	0.7246	0.24	0.01	10.24	84	0.7262	0.42	0	10.5	162	0.7254	0.3218	0	10.5	
IB Reputation (%)	78	79.75%	1	0	1	84	25.30%	0	0	1	162	51.85%	0	0	1	***
VC Backing (%)	78	50.63%	1	0	1	84	7.23%	1	0	1	162	28.40%	0	0	1	***
Age (years)	78	17.447	13.003	2.063	93	84	16.003	12.3671	0.8301	102.5863	162	16.707	12.9753	0.8301	102.5863	
Market Condition (%)	78	1.63%	2.01%	-13.22%	12.90%	84	0.79%	0.80%	-12.16%	12.90%	162	1.20%	1.95%	-13.22%	12.90%	

Table 3 Descriptive statistics of the key characteristics of IPO backed firms

Variable	Obs	Mean	Median	Min	Max
Total No. of Business Groups	78	8.167	7	1	31
Foreign Allocation (%)	78	53%	54%	0%	100%
Investment Banker Affiliation (%)	78	16.67%	0	0	1
Subs at Upper Limit of Band(%)	78	71.79%	1	0	1
Subs to Max. Limit (%)	78	84.62%	1	0	1
Oversubscription_Anchor (Times)	76	1.478	1.17	1	3.97

Table 4: Certificate Mechanisms and Institutional and Retail Subscription. The table reports estimates of OLS regression based on the total sample of 162 IPOs during July 2009-March 2016. The dependent variable is the natural logarithm of subscription. Model 1 and Model 2 uses natural logarithm of total subscription of both institutional (QIB) and retail investors (RI). Model 3 and Model 4 are based on natural logarithm of first day subscription of institutional and retail investors. Model 5 and 6 uses the demand build up of the natural logarithm of institutional and retail investors from first day to penultimate day. Model 7 and 8 are based on the natural logarithm of last day subscription of institutional and retail investors. Anchor investor is a dummy variable which takes a value of 1 if an IPO is anchor backed 0 otherwise. IB Reputation is a measure of the reputation of the investment banker and takes a value of 1 if there is a minimum of one investment banker in the top five ranks by market share of total IPO mobilization in the IPO year, and zero otherwise. VC backing is a dummy variable that takes a value of 1 if an IPO is VC backed and 0 otherwise. IPO Grade is the grade awarded to the IPO firm by a grading agency or 0 for non-graded IPOs. Age is the natural logarithm of the difference between the firm's IPO year and its incorporation year. Issue size is the natural logarithm of issue proceeds (in crs). Market condition is the Index return of thirty days before the IPO opens. Log_Penultimate RI Subscription is natural log of ratio of the total shares subscribed at the end of the penultimate day of book building to the total number of shares available to retail investors. Log_Penultimate QIB Subscription is natural log of ratio of the subscriptions by institutional investors at the end of the penultimate day of the book building process to the total number of shares available to them. The study control for year fixed effects using dummy variable. t-statistics based on robust standard errors are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels respectively.

	Total Subscription QIB	Total Subscription RI	First Day Subscription QIB	First Day Subscription RI	First to Penultimate Day Subscription QIB	First to Penultimate Day Subscription RI	Last Day Subscription QIB	Last Day Subscription RI
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intercept	-0.7656931	2.149641 ***	-0.2912141	0.3973291 **	-0.957345 *	-0.172379	0.2678821	2.069212 ***
	(-1.3844)	(5.5089)	(-1.345)	(3.2332)	(-1.7719)	(-0.606)	(0.3504)	(5.112)
Anchor Investors	0.7020715 ***	-0.289288 **	-0.1215356	-0.0637772	0.1457	-0.205115 *	0.6431782 **	-0.208732
	(2.591)	(-2.0386)	(-1.2605)	(-1.3939)	(0.5652)	(-1.7121)	(2.0623)	(-1.5486)
IB Reputation	0.2192621	-0.204462	0.0153788	-0.0925249	0.177839	-0.017061	0.0381895	-0.189785
	(0.8824)	(-1.311)	(0.1845)	(-1.4852)	(0.9663)	(-0.1589)	(0.1321)	(-1.2011)
VC Backing	0.0800277	-0.024735	-0.0111001	0.0089777	0.185302	0.122249	-0.0081973	-0.071967
	(0.3613)	(-0.1688)	(-0.1174)	(0.1615)	(0.9389)	(1.3107)	(-0.0309)	(-0.5088)
IPO Grade	0.2248647 **	0.184675 ***	0.0050743	0.0241047	0.144065 *	0.143358 **	0.1362034	0.185661 ***
	(2.0834)	(2.7213)	(0.1468)	(0.6056)	(1.7745)	(2.2765)	(1.2557)	(2.9068)
Log_Penultimate Subscription QIB		0.522451 ***						0.524185 ***
		(7.6389)						(8.0115)
Log_Penultimate Subscription RI	0.8329659 ***						0.4431773 **	
	(5.3592)						(2.0795)	
Log_Age	0.1307927	0.022491	0.0311504	-0.0066041	0.14188	0.088846	0.0645718	0.019279
	(1.2381)	(0.3368)	(0.5756)	(-0.2473)	(1.1787)	(1.5299)	(0.4704)	(0.2822)
Log_Issue Size	0.2084123 **	-0.328508 ***	0.1695986 ***	-0.0394556 **	0.141799 *	-0.025023	0.090526	-0.340858 ***
	(2.4497)	(-4.5985)	(3.9811)	(-2.3629)	(1.6899)	(-0.7073)	(0.8173)	(-4.7097)
Market Condition	2.2203875	1.313852	-0.0289611	0.1812191	0.907709	0.100784	2.563997 *	1.266113
	(1.7955) *	(1.1644)	(-0.0518)	(0.4091)	(0.7361)	(0.1483)	(1.7295)	(1.145)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of Observations	162	162	162	162	162	162	162	162
Adj R-squared	0.4897	0.4105	0.3632	0.08533	0.25	0.1749	0.3059	0.4207

Table 5: Certification mechanisms and underpricing. The table presents the OLS regression results of different certifications on IPO underpricing, on the full sample of 162 IPOs between July 2009 –March 2016. The dependent variable, underpricing is the market adjusted difference between the offer price & the first-day closing price of an IPO. Model 1 checks the effect of presence of anchors on underpricing. Model 2 considers all the certification mechanisms. Model 3 is based on all the certification mechanisms and total institutional subscriptions. Model 4 is based on all the certification mechanisms and total retail subscriptions. Model 5 is based on all the certification mechanisms and total subscriptions Anchor investor is a dummy variable which takes a value of 1 if an IPO is anchor backed 0 otherwise. IB Reputation is a measure of the reputation of the investment banker and takes a value of 1 if there is a minimum of one investment banker in the top five ranks by market share of total IPO mobilization in the IPO year, and zero otherwise. VC backing is a dummy variable that takes a value of 1 if an IPO is VC backed and 0 otherwise. IPO Grade is the grade awarded to the IPO firm by a grading agency or 0 for non-graded IPOs. Age is the natural logarithm of the difference between the firm's IPO year and its incorporation year. Issue size is the natural logarithm of issue proceeds (in crs). Market condition is the Index return of thirty days before the IPO opens. Log_Total Subscription RI is natural log of ratio of the total shares subscribed by retail investors to the total number of shares available to retail investors. Log_Total Subscription QIB is natural log of ratio of the subscriptions by institutional investors to the total number of shares available to them. Log_Total Subscription is natural log of ratio of the total subscriptions by all investors to the total number of shares. The study control for year fixed effects using dummy variable. t-statistics based on robust standard errors are in parentheses. ∗∗∗, ∗∗ and ∗ denote significance at the 1%, 5% and 10% levels respectively.

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**
Intercept	0.1229608	0.0542441	0.1153079	-0.1513865	-0.00820255
	(0.8544)	(0.3516)	(0.7868)	(-1.0026)	(-0.0586)
Anchor Investors	0.0257384	0.0412612	-0.0135136	0.0718902	0.02015745
	(0.4898)	(0.859)	(-0.2909)	(1.5538)	(0.4746)
IB Reputation		-0.1048704 *	-0.1221293 **	-0.0903879 *	-0.09023459 *
		(-1.7411)	(-2.3403)	(-1.6589)	(-1.7339)
VC Backing		-0.0244928	-0.0414437	-0.0323465	-0.04018068
		(-0.4949)	(-1.0618)	(-0.822)	(-1.0869)
IPO Grade		0.0422542 *	0.0071112	0.0094009	0.00086323
		(1.7251)	(0.2797)	(0.3948)	(0.0364)
Log_Total Subscription QIB			0.1071563 ***		
			(5.0873)		
Log_Total Subscription RI				0.1238199 ***	
				(3.8739)	
Log_Total Subscription					0.1144399 ***
					(6.0537)
Log_Age	-0.0039986	-0.0144611	-0.0347788	-0.0260472	-0.03645574
	(-0.1183)	(-0.4203)	(-1.0168)	(-0.7411)	(-1.0821)
Log_Issue Size	-0.0118261	-0.0066055	-0.0245407	0.0206229	-0.00633816
	(-0.5829)	(-0.2829)	(-1.1294)	(0.9957)	(-0.3113)
Market Condition	0.1162167	0.1558065	-0.1028225	-0.0792245	-0.18083504
	(0.2494)	(0.3257)	(-0.2214)	(-0.1733)	(-0.4029)
Year Dummies	Yes	Yes	Yes	Yes	Yes
No. of Observations	162	162	162	162	162
Adj R-squared	-0.0324	-0.02693	0.08003	0.05111	0.08811

Table 6: Anchor Characteristics and Subscription. The table reports estimates of OLS regression and the observations are based only for anchor-backed IPOs. During July 2009 –March 2016, anchors invested in 78 IPOs. The dependent variable is the natural logarithm of subscription of both institutional and retail investors. Model 1 and 4 checks the impact of anchors identity on subscription of intuitional and retail investors. Model 2 and 5 checks the impact of anchors investment decisions on subscription of intuitional and retail investors. Model 3 and 6 is based on all the anchors specific characteristics and their impact on subscription of intuitional and retail investors. Subscription QIB is natural log of ratio of total shares subscribed by the QIBs to the total number of shares available to them in the anchor backed IPOs and Subscription RI is natural log of the ratio of total shares subscribed by the retail investors to the total number of shares available to them in the anchor backed IPOs. Total No. of Business Group is the number of unique business groups participating as anchor investors in an IPO. Foreign Allocations are the percentage of total equity shares allocated to the foreign anchors. Investor Banker Affiliation is a dummy variable that takes a value of 1 if anchor investor(s) is affiliated to the investment banker(s) of the issue and 0 otherwise. Subscription at Upper Limit of Band is a dummy variable which takes a value of 1 if the allotment of shares to anchor investors is made at the upper end of the price band and 0 otherwise. Subscription to maximum limit is a dummy variable which takes a value of 1 if the anchor investors are allotted their entire reserved quota of shares and 0 otherwise. Oversubscription_Anchor is measured as the ratio of the total shares subscribed by the anchor investors to the total number of shares available to them. Age is the natural logarithm of the difference between the firm's IPO year and its incorporation year, Issue size is the natural logarithm of issue proceeds (in crs). Market condition is the Index return of thirty days before the IPO opens. The study control for year fixed effects using dummy variable. t-statistics based on robust standard errors are in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels respectively

	Subscription QIB	Subscription QIB	Subscription QIB	Subscription RI	Subscription RI	Subscription RI
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	3.87201214 ***	-3.015377 ***	-0.8495785	3.062755 ***	-0.467437	-0.0243426
	(4.5054)	(-3.154)	(-0.8772)	(3.4691)	(-0.6302)	(-0.0286)
Total No. of Business Groups	0.13325867 ***		0.0730936 **	0.040987 **		0.0046894
	(4.3141)		(2.6407)	(2.0593)		(0.2698)
Foreign Allocation	-0.84006153 ***		-0.6213005 **	-0.7104 **		-0.5163677 *
	(-3.119)		(-2.0203)	(-2.0699)		(-1.9989)
Investment Banker Affiliation	0.18685536		-0.0106862	0.461556 ***		0.4371295 ***
	(0.7187)		(-0.0328)	(2.9424)		(2.9047)
Subs at Upper Limit of Band		0.513994 **	0.3770533 **		0.311033	0.3659153
		(2.3379)	(2.0927)		(1.2576)	(1.3959)
Subs to Max. Limit		0.590523 *	0.5528399 *		0.44893 **	0.3817017 **
		(1.7887)	(1.7412)		(2.2455)	(2.2134)
Oversubscription_Anchor		0.918139 ***	0.6812364 ***		0.544267 ***	0.4985049 ***
		(5.7275)	(3.6426)		(4.5182)	(4.6323)
Log_Age	0.00046611	0.081271	-0.0024629	-0.03656	-0.010214	-0.0501092
	(0.0036)	(0.567)	(-0.021)	(-0.3635)	(-0.1141)	(-0.5607)
Log_Issue Size	-0.26120522 **	0.428697 ***	0.2079809 *	-0.293609 **	-0.027879	-0.0180317
	(-2.1855)	(4.1988)	(1.8703)	(-2.0223)	(-0.3182)	(-0.1684)
Market Condition	6.27533167 ***	5.582965 ***	5.1374688 ***	4.952605 **	3.444802	3.8085453 *
	(2.9577)	(3.4697)	(3.0552)	(2.0591)	(1.5713)	(1.7419)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
No. of Observations	78	76	76	78	76	76
Adj R-squared	0.3577	0.4338	0.4792	0.2311	0.3638	0.3765

Table 7: Anchor Characteristics and Underpricing. The table reports estimates of OLS regression and the observations are based only for anchor-backed IPOs. During July 2009 –March 2016, anchors invested in 78 IPOs. The dependent variable, underpricing is the market adjusted difference between the offer price and the first-day closing price of an IPO. Model 1 checks the impact of anchors identity on underpricing. Model 3 checks the impact of anchors investment decisions on underpricing. Model 3 is based on all the anchors specific characteristics and their impact on underpricing. Total No. of Business Group is the number of unique business groups participating as anchor investors in an IPO. Foreign Allocations are the percentage of total equity shares allocated to the foreign anchors. Investor Banker Affiliation is a dummy variable that takes a value of 1 if anchor investor(s) is affiliated to the investment banker(s) of the issue and 0 otherwise. Subscription at Upper Limit of Band is a dummy variable which takes a value of 1 if the allotment of shares to anchor investors is made at the upper end of the price band and 0 otherwise. Subscription to maximum limit is a dummy variable which takes a value of 1 if the anchor investors are allotted their entire reserved quota of shares and 0 otherwise. Oversubscription_Anchor is measured as the ratio of the total shares subscribed by the anchor investors to the total number of shares available to them. Age is the natural logarithm of the difference between the firm's IPO year and its incorporation year, Issue size is the natural logarithm of issue proceeds (in crs). Market condition is the Index return of thirty days before the IPO opens. The study control for year fixed effects using dummy variable. t-statistics based on robust standard errors are in parentheses. ∗∗∗, ∗∗ and ∗ denote significance at the 1%, 5% and 10% levels respectively.

	Model 1	Model 2	Model 3
Intercept	0.3733785	-0.724196 ***	-0.2474669
	(1.2889)	(-4.3397)	(-1.5061)
Total No. of Business Groups	0.0214734 ***		0.0156289 ***
	(3.6273)		(3.7388)
Foreign Allocation	-0.0990249		-0.0157172
	(-1.6458)		(-0.2902)
Investment Banker Affiliation	-0.0177055		0.0108641
	(-0.2997)		(0.1701)
Subs at Upper Limit of Band		0.0823839 **	0.0468761
		(2.4306)	(1.5982)
Subs to Max. Limit		0.0051327	-0.0041175
		(0.0864)	(-0.0795)
Oversubscription_Anchor		0.2124775 ***	0.1673595 ***
		(4.8304)	(3.592)
Log_Age	0.0050615	0.0270823	0.0116772
	(0.1868)	(1.0019)	(0.4741)
Log_Issue Size	-0.0615752	0.0448702 **	-0.0183777
	(-1.3649)	(2.4)	(-0.8602)
Market Condition	0.3891797	0.1408295	-0.0453132
	(0.7538)	(0.4275)	(-0.1255)
Year Dummies	Yes	Yes	Yes
No. of Observations	78	76	76
Adj R-squared	0.1335	0.3447	0.3873

Pearson's Correlation of the variables used for the study. ***, ** and * denote significance at the 1%, 5% and 10% levels respectively.

	LogTotalsubs QIB	LogTotal subs RI	LogAge	Market condition	Log Issue Size	Underpricing	Anchor Backed	IB Reputation	VC Backing	IPO Grading
LogTotalsubs QIB	1									
LogTotal subs RI	0.3766867***	1								
LogAge	0.1834202*	0.1138138	1							
Market condition	0.2088528**	0.1336188·	0.04473647	1						
Log Issue Size	0.5114086***	-0.2514563**	0.087592	0.1559863*	1					
Underpricing	0.2750019***	0.3144532***	0.009806376	0.04491955	-0.01254047	1				
Anchor Backed	0.4713886***	-0.201202**	0.03737556	0.07324948	0.4969178***	0.03245959	1			
IB Reputation	0.4584352***	-0.1666174*	0.03948112	0.1653196*	0.6752105***	-0.05267493	0.544664***	1		
VC Backing	0.2791152***	-0.06575628	0.0369844	0.04395072	0.2543332**	0.00247935	0.4811432***	0.2780398***	1	
IPO Grading	0.5939316***	0.1651039*	0.1521556·	0.1122089	0.2596502**	0.04285297	0.02017807	0.2263334**	0.2003559**	1